

Gillian D. Beltz-Mohrmann, Ph.D.

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Professional Appointments	Assistant Professor - Smith College	2025 –
	Department of Physics and Program in Statistical & Data Sciences	
	Postdoctoral Research Fellow - Argonne National Laboratory	2022 – 2025
	Cosmological Physics and Advanced Computing Group	
Education	Graduate Research Assistant - Vanderbilt University	2016 – 2022
	Department of Physics & Astronomy	
	Vanderbilt University , Nashville, TN	May 2022
	Ph.D. in Astrophysics	Advisor: Andreas Berlind
1st & 2nd Author Publications	Thesis: <i>Developing an Accurate Probe of the Galaxy-Halo Connection: Baryonic Effects, Small-Scale Galaxy Clustering, and Halo Model Extensions</i>	
	Wellesley College , Wellesley, MA	May 2016
	B.A., <i>cum laude</i>	Advisors: Kim McLeod, James Battat
	Major: Astrophysics; Minor: German	
Submitted & Published	Submitted & Published	
	Total Citations: 94	
	6. Beltz-Mohrmann, G. D. , Pope, A., et al., 2025, “Illuminating the Physics of Dark Energy with the Discovery Simulations,” <i>The Open Journal of Astrophysics</i> , 8, 74	
	5. Pearl, A. N., Beltz-Mohrmann, G. D. , Hearin, A. P., 2024, “DiffOpt: Parallel optimization of Jax models,” <i>Journal of Open Source Software</i> , 9(104), 7522	
	4. Beltz-Mohrmann, G. D. , Szewciw, A. O., Berlind, A. A., Sinha, M., 2023, “Toward Accurate Modeling of Galaxy Clustering on Small Scales: Halo Model Extensions and Lingering Tension,” <i>The Astrophysical Journal</i> , 948, 100	
	3. Szewciw, A. O., Beltz-Mohrmann, G. D. , Berlind, A. A., Sinha, M., 2021, “Toward Accurate Modeling of Galaxy Clustering on Small Scales: Constraining the Galaxy-Halo Connection with Optimal Statistics,” <i>The Astrophysical Journal</i> , 926, 15	
	2. Beltz-Mohrmann, G. D. , Berlind, A. A., 2021, “The impact of baryonic physics on the abundance, clustering, and concentration of halos,” <i>The Astrophysical Journal</i> , 921, 112	
	1. Beltz-Mohrmann, G. D. , Berlind, A. A., Szewciw, A. O., 2020, “Testing the Accuracy of Halo Occupation Distribution Modelling using Hydrodynamical Simulations,” <i>Monthly Notices of the Royal Astronomical Society</i> , 491, 5771	
	In Prep.	
	2. Beltz-Mohrmann, G. D. et al., “Introducing DiffSky: A Differentiable Model of the Galaxy-Halo Connection”	
Nth Author Publications	1. Beltz-Mohrmann, G. D. et al., “The DESI Emulator Mock Challenge”	
	Submitted & Published	
	Total Citations: 92	
	8. Baumer, B. S. et al., 2026, “Emerging Models for a Second Undergraduate Course in Data Science”, submitted to <i>Harvard Data Science Review</i>	
	7. Lange, Johannes U. et al., 2025, “Cosmological Constraints from Full-Scale Clustering and Galaxy-Galaxy Lensing with DESI DR1”, <i>arXiv:2512.15962</i>	

6. Alarcon, Alex et al., 2025, “DiffstarPop: A generative physical model of galaxy star formation history”, submitted to The Open Journal of Astrophysics
5. OpenUniverse Collaboration et al., 2025, “OpenUniverse2024: A shared, simulated view of the sky for the next generation of cosmological surveys”, Monthly Notices of the Royal Astronomical Society
4. Lange, Johannes U. et al., 2024, “Systematic Effects in Galaxy-Galaxy Lensing with DESI”, The Open Journal of Astrophysics, 7, 57
3. Yuan, Sihan et al., 2024, “Redshift evolution and covariances for joint lensing and clustering studies with DESI Y1”, Monthly Notices of the Royal Astronomical Society, 533, 1
2. Dale, D. A. et al., 2016, “Radial Star Formation Histories in Fifteen Nearby Galaxies,” The Astronomical Journal, 151, 4
1. Souza, S. P., Beltz-Mohrmann, G., Sami, M., 2014, “The Light Curve and Period of MT696,” The Journal of the American Association of Variable Star Observers, 42, 154

In Prep.

2. Paillas, E. et al., “Simulation-Based Emulators for Galaxy Clustering in the Era of Stage-IV Surveys: I. Two-Point Statistics and Beyond”
1. Kraskov, I. et al., “The Impact of Inter-Galactic Medium Absorption Model Misspecification on Photometric Redshifts”

Teaching

Smith College

PHY 211 Computational Method in the Physical Sciences	Spring '26
PHY 210 Mathematical Methods of Physical Sciences & Engineering	Spring '26
SDS 271 Programming for Data Science in Python	Fall '25, '26
SDS 291 Multiple Regression	Fall '25, '26

Conference for Undergraduate Women in Physics, ANL

Python workshop	Jan. 2023
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Vanderbilt University

Astronomy Lab Instructor	Fall 2016 – Spring 2019
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Wellesley College

Supplemental Instruction Leader	Fall 2014 – Spring 2016
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Mentoring

Madeline Dean , Smith College	Fall 2025 –
Topic: Modeling Baryonification with Jax	
Topic: Modeling galaxies with HODs in Λ CDM and w_0w_a CDM simulations	
Emma Schultz , Smith College	Spring 2026 –
Topic: Modeling IGM attenuation with Jax	
Zhiwen Ji , Smith College	Fall 2025
Topic: Investigating the concentration-halo mass relation in IllutrisTNG	
Harmandeep Gill , University of Toronto (undergraduate)	Oct. 2024 – May 2025
Topic: Modeling Lyman Break Galaxies	
Ivan Kraskov , University of Toronto (undergraduate)	Oct. 2024 – Oct. 2025
Topic: Modeling IGM attenuation in Jax	
Emily Martsen , University of Chicago (graduate)	Sep. 2024 – June 2025
Topic: Measuring the Two-point Clustering of Galaxy Clusters	
Resherle Verna , UT Austin (graduate), GEM Fellowship Program	Summer 2023
Topic: Forward modeling galaxy SEDs with Jax	
Caleigh Dennis , Harpeth Hall High School	Sep. 2017 – May 2019
Topic: Measuring the rotation of galaxy groups in SDSS	
1st place winner at Middle Tennessee Science & Engineering Fair (2018 & 2019)	

Recent Talks	New Frontiers in Cosmology , A Coruña, Spain	August 2026
	<i>Modeling the Galaxy-Halo Connection with Diffsky</i>	
	Summer DESI Meeting , Durham, UK	July 2026
	<i>Using Alternative Clustering Methods to Constrain Cosmology</i>	
	AAS 248 , Pasadena, CA	June 2026
	<i>Results from the DESI Emulator Mock Challenge</i>	
	Five College Astronomy Department Colloquium , UMass Amherst	Feb. 2026
	<i>Improving Models of Small-Scale Galaxy Clustering to Constrain Cosmology</i>	
	Mock NYC , Center for Computational Astrophysics	Jan. 2026
	<i>The DESI Emulator Mock Challenge</i>	
	Cosmology from Lyman-Break Galaxies , University of Toronto	May 2025
	<i>Modeling the Galaxy-Halo Connection of LBGs</i>	
	ELG Mock Challenge , Donostia International Physics Center	Feb. 2025
	<i>Lessons Learned from the DESI Emulator Mock Challenge,</i>	
	<i>Simulation-based Forward Modeling with Diffsky</i>	
	Winter DESI Meeting , Cancun, Mexico	December 2024
	<i>Updates on the DESI Emulator Mock Challenge</i>	
	Taco Talk , University of Arizona	October 2024
	<i>A Differentiable Forward Model of the Galaxy-Halo Connection</i>	
	Cosmology Talks Miniworkshop (invited expert)	August 2024
	<i>Cosmology Beyond 2pt Statistics</i>	
	DHWFEST , University of Utah	July 2024
	<i>A New Forward Model of the Galaxy-Halo Connection</i>	
	Summer DESI Meeting , Marseille, France	July 2024
	<i>DESI Alternative Clustering Methods</i>	
	New Strategies for Extracting Cosmology from Galaxy Surveys	July 2024
	Sexten Center for Astrophysics	
	<i>Simulation-based Forward Modeling of Cross-Survey Cross-Correlations with Diffsky</i>	
	Fundamental Physics from Future Spectroscopic Surveys , LBNL	May 2024
	<i>Making multi-wavelength, multi-redshift predictions for Cross-Survey Cosmological Analyses</i>	
	Winter DESI Meeting , Hawaii, USA	Dec. 2023
	<i>Introducing DESI-Diffsky: A Differentiable Forward Model for Making</i>	
	<i>Multi-wavelength, Multi-tracer DESI Mocks</i>	
	Building a physical understanding of galaxy evolution with data-driven astronomy , KITP, UC Santa Barbara	Jan. 2023
	<i>Toward Accurate Modeling of Galaxy Clustering on Small Scales:</i>	
	<i>Halo Model Extensions & Lingering Tension</i>	
	CAMELS Workshop , Center for Computational Astrophysics	Dec. 2022
	<i>Toward Accurate Modeling of Galaxy Clustering on Small Scales:</i>	
	<i>Halo Model Extensions & Lingering Tension</i>	
	N-Body Shop Workshop , Center for Computational Astrophysics	June 2022
	<i>Accurate Modeling of Galaxy Clustering on Small Scales</i>	
	High-Energy and AstroPhysics Seminar , University of Utah	Jan. 2022
	<i>Developing an Accurate Probe of the Galaxy-Halo Connection</i>	
	KICP Seminar , University of Chicago	Nov. 2021
	<i>Developing an Accurate Probe of the Galaxy-Halo Connection</i>	
	Galaxy Lunch talk , Yale University	March 2021
	<i>Can we ignore baryons in halo modeling?</i>	
	Galaxy-Halo Connection Across Cosmic Time , UC Santa Barbara	Aug. 2020
	<i>HMF Discrepancies between Hydrodynamic and DMO Simulations</i>	
	Mock Innsbruck (remote participant)	March 2020
	<i>Taking Halo Modeling to the Next Level</i>	

In the Media

DESI Blog: At the Big Reveal: DESI's December 2024 Unblinding Results

Podcast: Particle Mysteries: The Coldest Case - "Chasing Shadows"

YouTube: Science 101: What are dark matter and dark energy?

Grants	XSEDE Grant	2019, 2020
	Awarded 58.4k total Node Hours (2.8M CPU hours) on Stampede2	
	McMinn Research Grant	2019, 2020
	Vanderbilt Physics & Astronomy Department (\$3,000 total)	
	Graduate Summer Research Award	2018
	Vanderbilt College of Arts and Sciences (\$1,900)	
Honors & Awards	Most Outstanding Student Publication Award	2020
	Vanderbilt Physics & Astronomy Department	
	Graduate Student Poster Competition Winner	2019
	Vanderbilt Data Science Symposium	
	Akunuri V. Ramayya Award for Outstanding Teaching Assistant	2018
	Vanderbilt Physics & Astronomy Department	
	Provost Graduate Fellowship	2016 – 2021
	Vanderbilt University	
	Undergraduate Chambliss Achievement Honorable Mention	2016
	227th American Astronomical Society Meeting	
	Sarah Frances Whiting Medal for Achievement in Astronomy	2014
	Wellesley College	
Skills & Experience	Programming Languages: PYTHON, C, BASH, GIT, L ^A T _E X	
	Misc.: Jax, scikit-learn, emcee, GADGET, CAMB, 2LPTIC, ROCKSTAR	
	Parallel Computing: MPI, OPENMP	
Public Service & Outreach	Future Physicists – 5 College Women+ Career Day, Smith College	Jan. 2026
	DESI Mentorship Program	Oct. 2023 –
	Science Careers in Search of Women panelist, Argonne	March 2023
	Conference for Undergraduate Women in Physics, Argonne	Jan. 2023
	AAS Congressional Visits Day (virtual)	Sept. 2020
	Science Day with Nashville Girl Scout Troop	March 2019
	Meet the Astronomer Night at Dyer Observatory	Oct. 2018
	Vanderbilt Student Volunteers for Science	Fall 2016
	Whitin Observatory Volunteer, Wellesley College	2012 – 2016
Collaborations	Dark Energy Spectroscopic Instrument (DESI)	2022 –
	C3 and GQC Working Groups	
	Alternative Clustering Methods Topical Group (co-leader, November 2024 –)	
	LSST Dark Energy Science (DESC)	2022 –
	Large Suite of Dark Matter Simulations (LasDamas)	2017 – 2022
	Co-Investigator & XSEDE Allocation Manager	
Professional Service	American Astronomical Society	2015 –
	NSF Review Panel	April 2026
	Scholarly Journal Peer Reviewer	
	Monthly Notices of the Royal Astronomical Society	
	Astronomy & Astrophysics	
	Journal of Cosmology and Astroparticle Physics	
	Physics of the Dark Universe	